Answers and talk notes

Explore Integers and Their Opposites

What your student is learning

Tonight we write integers for real situations — gains and losses, above and below — and say what ZERO means in each one.

Answers

Ask your student to explain before you look.

1. **B.** –15. A loss is negative, so the change is –15.
2. **A.** –27. The opposite of 27 is –27. Both are 27 units from 0.
3. **D.** 0. A gain of 9 and a loss of 9 cancel, so the total change is 0.
4. **C.** 5 feet below the ground. Below the ground is negative, so 5 feet below the ground is –5.
5. **B.** –33. The first opposite is 33, and the opposite of 33 is –33. Two flips return to the start.
6. **A.** –30. Under the ground is below 0, so the cave is –30.

Talk about it

After two flips across 0, where do you end up? What does that tell you?

Home mission

Write –58 on one side of a paper and 58 on the other. Flip it three times. Can your student say what shows, and why?

When they are stuck, ask

- How far from zero is it, and which side is it on?
- What does zero mean in this story?
- Is this a gain or a loss? Up or down?

Try not to say

- The opposite of a negative is still negative.
- Negative numbers are just numbers with a minus in front, ignore it.
- –12 is bigger than –3 because 12 is bigger.